

Flight MH370 Revisited

Large, P.O., Large, J.T., Sarbin, M., _____,
[Affiliation(s)]
[Corresponding author email and ORCID]

Abstract

[Write the abstract as a single self-contained paragraph of approximately 150 words maximum. Briefly state the problem addressed, the purpose of revisiting Flight MH370, the scope of the meta-analysis/reproduction, the additional observables examined, the principal results, and the main conclusion/recommendation. Avoid citations and undefined abbreviations.]

[Keywords are not included in the manuscript. Select up to four in the online submission system.]

1. INTRODUCTION

[Set out the unresolved navigation/search problem, the relevant history of the MH370 analysis, the motivation for revisiting the evidence, and the contribution of this paper. Define all abbreviations at first use and clearly distinguish established facts, prior published results, reproduced results, and new analysis.]

Malaysian Airlines Flight 370, a Boeing 777-200ER registered in Malaysia as 9M-MRO, was lost on March 7th, 2014 with 239 souls on board. At this time, more than twelve years after the loss, neither the flight recorders nor the wreckage of the aircraft have been located despite an extensive search, with the exception of multiple items of debris recovered from island beaches in the Indian Ocean and along the south-eastern African coast and confirmed to be from the accident aircraft. The specific events relating to the flight, insofar as they are known, are well documented in the official reports of the safety investigation (ATSB, 2017; Malaysian Government, 2018).

The Safety Investigation Report (Malaysian Government, 2018), while noting many important facts and making multiple recommendations, concluded that the investigation was “unable to determine the real cause for the disappearance of MH370”. This conclusion has significance for aviation safety, given the importance of understanding the exact circumstances and probable cause of a hull loss of this magnitude and significance, and in order to have confidence in the efficacy of recommendations for future preventative action.

An extensive subsea search effort – the largest of its kind in history - covering approximately 232,000km² (update to 2026) of the ocean floor (Malaysian Government, 2018 add citation for Ocean Infinity search area) with high resolution sonar data, based on geospatial parameter estimates derived using the available information, has proven unsuccessful at the time of writing.

The official search area for a large part of the search was defined with reference to a posterior probability density function estimated by Davey, Gordon, Holland, Rutten and Williams (2015), where the posterior probability density estimates reflect the estimated likelihood of the aircraft state vector taking a particular value given the available observational data and its inherent uncertainty, constrained by plausible flight dynamics. Davey et al. (2015) state that “The factors that do make a significant difference to the output pdf are the assumed spread of Mach number and the end of flight model” (p.98). The Mach number was sampled between

M0.73 and M0.84 and the end of flight model assumed no human control, which together have a significant impact on the search area. The dynamics at the end of the flight, i.e. at 00:19 UTC and immediately afterwards and the subsequent assumptions about the flight trajectory at that time were partly informed by Holland (2017?) (ATSB___ did they actually reference this?)

Fundamentally, there are a limited number of possible explanations for why the search thus far has been unsuccessful. Either: (a) The searches to date have covered the area where the wreckage being sought exists, but detection was not successful. The confidence in the ability of the subsea search to locate the debris field if contained within the search is assessed as $p=0.94$ (ATSB, 2017). That is, it is possible for the ocean floor survey to miss the debris field due to small gaps in the data, areas of terrain shadowing on the ocean floor, or the failure of the means of detection to identify the debris. Or, (b) The wreckage is located in an area outside the higher probability region as originally estimated, including the end of flight assumptions, but still within the estimated probability density function, or (c) The area of the estimated posterior probability density function above a nominal probability threshold, and including the end of flight assumptions, does not in fact contain the aircraft wreckage.

For the purposes of the present study, the available useful information includes network management and signalling data from log files pertaining to the incident aircraft's L-band satellite communications system, aircraft performance information for the aircraft type in question combined with relevant data such as field on board, the location of items of debris confirmed to either be from the specific aircraft in question or having been determined to be highly likely to be so, which can be combined with ocean drift models, and other related information as noted below.

L-Band data

2. META-ANALYSIS AND REPRODUCTION

[Describe the prior analyses included in the meta-analysis, inclusion/exclusion criteria, source data, assumptions and reproducibility approach. Explain how published results were reconstructed or independently reproduced, including any normalization or reconciliation of differing methods.]

3. ANTENNA GAIN AND RECEIVED POWER OBSERVABLE

[Develop the antenna-gain / received-power observable and its relationship to the available satellite communications data. State the physical model, calibration assumptions, uncertainties, and limitations. Separate reproduced methodology from any new method introduced here.]

4. ON THE END OF FLIGHT 370

[Examine the evidence and hypotheses concerning the final phase of flight. Clearly state which inferences follow from the observables, which depend on modelling assumptions, and which remain speculative or underdetermined.]

5. RESULTS

[Present the combined results of the reproduction, meta-analysis and additional observable(s). Where appropriate, quantify uncertainty and sensitivity to assumptions. Place figures and tables close to their first discussion in the text.]

6. CONCLUSIONS AND RECOMMENDATIONS

[Summarise what the analysis establishes, what it does not establish, and how the findings change or refine the current understanding of MH370. State any recommendations for future analysis, data release, search strategy, or investigation that follow directly from the results.]

ACKNOWLEDGEMENTS

[Optional. Acknowledge assistance, data access, funding and other contributions as appropriate.]

DATA AVAILABILITY STATEMENT

[State where the data and analysis materials needed to reproduce the work can be obtained, subject to any restrictions. If code, derived datasets or supplementary calculations are deposited in a repository, provide the citation/link at submission.]

REFERENCES

[Insert references in the Journal of Navigation style. Check every citation against the reference list.]

[Author surname], [Initials]. ([Year]). [Article title]. The Journal of Navigation, [volume], [pages].

MH370 MANUSCRIPT WORKING CHECKLIST — DELETE THIS PAGE BEFORE SUBMISSION

- ☐ Keep the abstract to one paragraph and approximately 150 words maximum.
- ☐ Use the supplied numbered section headings manually; add subsection numbers manually if needed (e.g. 2.1., 2.2.).
- ☐ Maintain a clear distinction between previously published findings, reproduced findings and new findings.
- ☐ For each reproduced result, identify the original source, input data, assumptions and any deviations from the original method.
- ☐ For the antenna gain / received power analysis, document calibration, geometry, uncertainty and sensitivity assumptions sufficiently for independent reproduction.
- ☐ For claims concerning the end of flight, label model-dependent inference and uncertainty explicitly.
- ☐ Report numerical uncertainty consistently and avoid more significant figures than the data support.
- ☐ Number Figures and Tables manually from 1; table captions above tables, figure captions below figures.
- ☐ Embed figures and tables in the initial manuscript near their first discussion.
- ☐ Decode every acronym at first use, including BTO, BFO, SDU, SATCOM and any Inmarsat terminology used.
- ☐ Use UK English throughout.
- ☐ Check that conclusions and recommendations follow directly from the presented analysis rather than introducing new evidence.
- ☐ Include a data availability statement and, where possible, make code/derived data available for reproducibility.
- ☐ Before submission, check the current Cambridge University Press Journal of Navigation author instructions for any changed requirements.

Although these exchanges do not include any normal ‘payload’ data such as the routine Aircraft Communications Addressing and Reporting System (ACARS) messages which would normally be expected during the flight, nor were two satellite telephony attempts successful, these network management and signaling exchanges provide a set of measurements useful for making spatio-temporal inferences about the flight after 18:22Z, beyond the most striking fact that the flight continued for almost five hours after the final primary radar contact when the aircraft was located between the Malacca Strait and the Andaman Sea, at which time the aircraft had already been airborne for 104 minutes. 5

As was first described by Ashton, Schuster-Bruce, College and Dickinson (2014), the log files from the satellite communication network contain, among other things, timing offset and frequency offset measurements which can be combined with other known information, such as the dynamics of the Inmarsat spacecraft with which the AES was communicating and details of algorithms used in the GES and AES systems, to make inferences relating to aircraft location, heading and velocity at the times that the measurements were made. When combined with known aircraft performance characteristics and limitations, feasible aircraft dynamics, and the known fuel-on-board (FOB) the aircraft at departure and upon reaching cruise altitude (via ACARS messages), these inferences have been extended to geospatial

estimation for the purposes of search area definition for the location of the aircraft wreckage, including the flight recorders and other physical evidence. Following recovery of multiple items of debris, either confirmed to be from the aircraft in question or determined as likely to be from that aircraft, reverse ocean surface drift models based on time of debris arrival at the point of discovery, combined with knowledge and estimation of ocean surface or near-surface dynamics and windage factors, have yielded additional geospatial estimates for the likely origin of the debris. Satellite imagery has also been made available which could contain objects from the aircraft, although not which have been positively identified as such.

$p=0.94$ (ATSB, 2017). That is, it is possible for the ocean floor survey to miss the debris field due to small gaps in the data, areas of terrain shadowing on the ocean floor, or the failure of the means of detection to identify the debris. determined, to a level of confidence reported as $p=0.94$ (, that the wreckage was not in the area in which the posterior probability density function from the Bayesian methods estimation predicted it would be, to a reported $p=0.85$ to 0.90 level of confidence (ATSB, 2017). Either, (a) The wreckage is located in the area outside the $p=0.85-0.90$ region but still within the estimated PDF i.e. in a **remaining $p=0.10-0.15$ area which was not searched (likely searched since)**, (b) if the defined search area and hence the $p=0.85-0.90$ pdf area did include the wreckage but it was not detected by the survey, for which the probability of this occurring is estimated to be $p=0.06$; or, (c) the area of the estimated posterior PDF does not contain the aircraft wreckage.

Of particular note is that the combined area searched covers, and in fact extends beyond, the area from the posterior probability density function described by the ATSB (2017) to contain the aircraft debris with an estimated probability in the range of $p=0.85-0.90$. The confidence in the ability of the subsea search to locate the debris field if contained within the search is assessed as $p=0.94$ (ATSB, 2017). That is, it is possible for the ocean floor survey to miss the debris field due to small gaps in the data, areas of terrain shadowing on the ocean floor, or the failure of the means of detection to identify the debris.

Fundamentally, there are a limited number of possible explanations for why the search determined, to a level of confidence reported as $p=0.94$, that the wreckage was not in the area in which the posterior probability density function from the Bayesian methods estimation predicted it would be, to a reported $p=0.85$ to 0.90 level of confidence (ATSB, 2017). Either, (a) The wreckage is located in the area outside the $p=0.85-0.90$ region but still within the estimated PDF i.e. in a **remaining $p=0.10-0.15$ area which was not searched (likely searched since)**, (b) if the defined search area and hence the $p=0.85-0.90$ pdf area did include the wreckage but it was not detected by the survey, for which the probability of this occurring is estimated to be $p=0.06$; or, (c) the area of the estimated posterior PDF does not contain the aircraft wreckage.

The latter

The initial search for the aircraft was in the South China Sea, given that normal aircraft surveillance and communications ceased at the point of handover from the Malaysian to Vietnamese Flight Information Region near waypoint IGARI, at 17:21 UTC, approximately 40 minutes after departure from Kuala Lumpur toward Beijing. However, subsequent forensic examination of civil and military primary radar data revealed partially complete information about the trajectory of the aircraft for approximately one hour after this initial deviation from the flight plan, for the phase of flight between 17:21 and 18:22 UTC. During this time, the aircraft departed entirely from the intended flight plan, crossing over Malaysia and broadly following airways above the Strait of Malacca.

After that time, for the phase of flight between 18:25 and 00:19 UTC the next morning, it was subsequently found from satellite communications log files that intermittent exchanges were recorded between the aircraft's L-band mobile satellite service (MSS) terminal and the Inmarsat communications network, via the I-3 F1 spacecraft located in a geosynchronous equatorial orbit nominally stationed above the equator at 64.5E longitude and received at the Inmarsat Ground Earth Station (GES) in Perth, Western Australia, as was first described by Ashton, Schuster-Bruce, College and Dickinson (2014),

Although these log file exchanges do not include normal position report information, they contain network management and signaling data which have been used to make spatio-temporal inferences about the flight after 18:22 UTC, beyond the most striking fact that the flight continued for almost five hours after the final primary radar contact when the aircraft was located between the Malacca Strait and the Andaman Sea, at which time the aircraft had already been airborne for 104 minutes.

When combined with known aircraft performance characteristics and limitations, feasible aircraft dynamics, and the known fuel-on-board the aircraft at departure and upon reaching cruise altitude (via routine ACARS messages), these inferences have been extended to geospatial estimation for the purposes of search area definition for the location of the aircraft wreckage, including the flight recorders and other physical evidence.

Following recovery of multiple items of debris, either confirmed to be from the aircraft in question or determined as likely to be from that aircraft, reverse ocean surface drift models based on time of debris arrival at the point of discovery, combined with knowledge and estimation of ocean surface or near-surface dynamics and windage factors, have yielded additional geospatial estimates for the likely origin of the debris. Satellite imagery has also been made available which could contain objects from the aircraft, although not which have been positively identified as such.

case includes the possibility that the wreckage is on or close to a different part of the 7th BTO arc to that searched, or elsewhere. This could be due to a combination of factors, for example the assumptions and techniques used in the dynamic model, both during the flight and at the end of the flight, the presence of biases in the observations used, the characterization of the stochastic properties of the observations, etc. Davey et al. (2015) state that "The factors that do make a significant difference to the output pdf are the assumed spread of Mach number and the end of flight model" (p.98). The Mach number was sampled between M0.73 and M0.84 and the end of flight model assumed no human control, which together have a

significant impact on the search area due to their narrowing effect, both along the arcs and orthogonally to the 7th arc.